Practice Problem Set 4

ECON 320 – Introduction to Mathematical Economics

Fall 2025

Optimization of Multivariable Functions (Chapter 11)

1. Define the following terms and explain their roles in optimization:

   1. Critical point
   2. Gradient vector
   3. Hessian matrix
   4. Definiteness of a matrix

2. For a twice-differentiable function $f(x, y)$, explain how the signs of f_{xx}, f_{yy}, and the determinant of the Hessian

$$H = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

determine whether a stationary point is a maximum, minimum, or saddle point.

3. Find and classify all critical points of the function

$$f(x, y) = 4x^2 + y^2 - 4xy + 2x - 3y.$$

4. Determine the local extrema (if any) of

$$f(x, y) = 3x^2y - y^3 - 12x^2 + 12y.$$

Compute and interpret the determinant of the Hessian.

5. Let $f(x, y, z) = 2x^2 + y^2 + z^2 - 2xy - 4xz + 6yz - 3x + y + 2z$.

   1. Find all critical points.

2. Evaluate the definiteness of the Hessian at those points.
3. Classify each as a local maximum, minimum, or saddle.

6. A firm's production function is $Q = x^{0.5}y^{0.5}$.

   1. Derive the marginal products MP_x and MP_y.
   2. Show that $f(x, y)$ exhibits diminishing marginal returns.
   3. Determine the nature of returns to scale.

7. A monopolist faces a demand function $p = 100-2Q$ and cost function $C(Q) = 10Q+0.5Q^2$.

   1. Express profit $\pi(Q)$ and find the profit-maximizing output.
   2. Verify second-order conditions using $d^2\pi/dQ^2$.
   3. Extend this to two products: $\pi(x, y) = 100x + 80y - x^2 - y^2 - xy - 20(x + y)$. Find and classify the stationary point.

Optimization with Equality Constraints (Chapter 12)

8. Explain the role of the Lagrange multiplier in economic optimization. What does it represent intuitively in problems of:

   1. utility maximization subject to a budget constraint, and
   2. cost minimization subject to a production requirement?

9. State the second-order sufficient conditions for a constrained extremum using the bordered Hessian. Explain why the factor $(-1)^m$ (where m is the number of constraints) appears in the test.

10. Maximize $f(x, y) = xy$ subject to $x + y = 10$.

1. Form the Lagrangian and find the stationary point.
2. Determine whether it is a maximum or minimum using the bordered Hessian.

11. A firm produces output using two inputs, x and y, with the production function

$$Q = 4x^{1/2}y^{1/2}.$$

The firm faces the cost constraint $C = 2x + y = 100$.

1. Set up the Lagrangian and solve for x, y, λ.
2. Compute the bordered Hessian and verify whether the solution minimizes cost.
3. Interpret the Lagrange multiplier economically.

12. Find the stationary points of

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to the constraint $x+y+z = 3$. Determine whether the solution represents a minimum or maximum.

13. Using the method of Lagrange multipliers, find the maximum and minimum values of

$$f(x, y) = x^2 + y^2$$

subject to $x + 2y = 4$.

14. Maximize the utility function $U(x, y) = x^{0.4}y^{0.6}$ subject to the budget constraint $4x+2y = 100$.

1. Derive the first-order conditions.
2. Find the optimal consumption bundle.
3. Verify the nature of the extremum using the bordered Hessian.